mergencystretch

Chapter 3. Do-Calculus and Algorithmic Identification

Rules of do-calculus, completeness, c-components, the ID algorithm

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3.1 Do-calculus

Motivation. The central question is whether there is a systematic way to decide if $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ can be expressed in terms of the observational distribution $P(\mathbf{v})$. The truncated factorization sometimes yields identifiable effects (back-door, front-door) and sometimes not (the bow graph). Do-calculus (Pearl 1995) supplies three graphical inference rules, adding/removing observations, exchanging actions and observations, adding/removing actions, which are *complete*: if an effect is identifiable, do-calculus derives it.

3.1.1 Three examples via the truncated factorization

Motivation. See how identification works, and fails, when the $\mathbf{U}$-terms of a semi-Markovian model must be eliminated by hand.

Theorem 3.1 (Generalized truncated factorization)

In a semi-Markovian model $\mathcal{M}$, the distribution generated by $\mathrm{do}(\mathbf{X} = \mathbf{x})$ is

$$P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}),$$
$$\mathbf{V}' = \mathbf{V} \setminus \mathbf{X},$$

and for $\mathbf{Y} \subseteq \mathbf{V} \setminus \mathbf{X}$, $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} \sum_{\mathbf{u}} \prod_{V_i \in \mathbf{V} \setminus \mathbf{X}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$.

Proof [Supplement]. Fix $\mathbf{u}$. In $\mathcal{M}_{\mathbf{x}}$ the mechanisms of $\mathbf{X}$ are replaced by constants and the others are unchanged, so $\mathbf{V}'$ is a deterministic function of $(\mathbf{x}, \mathbf{u})$ and, equivalently, $P(\mathbf{v}' \mid \mathrm{do}(\mathbf{x}), \mathbf{u}) = \prod_{V_i \in \mathbf{V}'} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$ with the factors of the original model (each $P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i)$ is the indicator that $f_i(\mathbf{pa}_i, \mathbf{u}_i) = v_i$; acyclicity makes the product a valid joint law obtained by solving the equations in topological order). The intervention does not change $P(\mathbf{u})$, so averaging over $\mathbf{u}$ gives the formula. ■

Identification succeeds when the right-hand side can be rewritten without $\mathbf{U}$.

3.1.1.1 Back-door graph

$Z \to X$, $Z \to Y$, $X \to Y$ with independent U_Z, U_X, U_Y. In $\mathcal{M}_x$: $Z \leftarrow f_Z(u_Z)$, $X \leftarrow x$, $Y \leftarrow f_Y(x, z, u_Y)$.

$$\begin{aligned} P(\mathbf{v}' \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) \\ &= \Big(\sum_{u_Z} P(z \mid u_Z) P(u_Z)\Big)\Big(\sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y)\Big)\Big(\sum_{u_X} P(u_X)\Big) \\ &= P(z) \sum_{u_Y} P(y \mid x, z, u_Y) P(u_Y \mid x, z) = P(z) P(y \mid x, z), \end{aligned}$$

using $U_Y \perp\!\!\!\perp \{X, Z\}$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z) P(z)$.

3.1.1.2 Bow graph

$X \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Y \leftarrow f_Y(x, u, u_Y)$.

$$P(y \mid \mathrm{do}(x)) = \sum_{u} P(u) \sum_{u_Y} P(y \mid x, u, u_Y) P(u_Y) \sum_{u_X} P(u_X) = \sum_{u} P(y \mid x, u) P(u).$$

U cannot be eliminated: $P(u)$ and $P(y \mid x, u)$ are not functions of $P(x, y)$. (Observationally, $P(\mathbf{v}) = \sum_u P(x \mid u, u_X) P(y \mid x, u, u_Y) P(\mathbf{u})$ weights u by $P(u \mid x)$ instead.)

3.1.1.3 Front-door graph

$X \to Z \to Y$, $X \leftrightarrow Y$ through U. In $\mathcal{M}_x$: $X \leftarrow x$, $Z \leftarrow f_Z(x, u_Z)$, $Y \leftarrow f_Y(z, u, u_Y)$.

$$
\begin{aligned}
P(z, y \mid \mathrm{do}(x)) &= \sum_{\mathbf{u}} P(z \mid x, u_Z) P(y \mid z, u, u_Y) P(\mathbf{u}) \\
&= \Big(\sum_{u_Z} P(z \mid x, u_Z) P(u_Z) \Big) \Big(\sum_{u, u_Y} P(y \mid z, u, u_Y) P(u, u_Y) \Big) \\
&= P(z \mid x) \sum_{u} P(y \mid z, u) \sum_{x'} P(u \mid x') P(x') \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x') P(x') && (Y \perp\!\!\!\perp X \mid Z, U) \\
&= P(z \mid x) \sum_{u, x'} P(y \mid z, u, x') P(u \mid x', z) P(x') && (U \perp\!\!\!\perp Z \mid X) \\
&= P(z \mid x) \sum_{x'} \sum_{u} P(y, u \mid z, x') P(x') = P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').
\end{aligned}
$$

Therefore the **front-door formula**

$$
P(y \mid \mathrm{do}(x)) = \sum_{z} P(z \mid x) \sum_{x'} P(y \mid z, x') P(x').
$$

3.1.2 Identification approaches

Motivation. State the syntactic goal that do-calculus serves.

In both the back-door and front-door settings the goal was to reduce $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ to an expression whose terms are estimable from $P(\mathbf{v})$. One way is to write the truncated factorization and derive a $\mathbf{U}$-free expression, as above. Another is to rewrite the query into a *do-free* expression without ever mentioning $\mathbf{U}$. We seek rules (axioms) that transform do-expressions into do-free ones while preserving equality with the target effect.

3.1.3 Do-calculus

Motivation. Derive the three rules from three insights and explain their graphical side conditions.

3.1.3.1 Prelude: independence in a submodel

$\mathrm{do}(\mathbf{X} = \mathbf{x})$ replaces f_X by the constant x for each $X \in \mathbf{X}$. Graphically $\mathcal{G}_{\overline{\mathbf{X}}}$ represents $\mathbf{X}$ not influenced by any other variable. Moreover $\mathbf{X}$ are now constants, which corresponds to conditioning on $\mathbf{X} = \mathbf{x}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. Hence

$$
(\mathbf{A} \perp\!\!\!\perp \mathbf{B} \mid \mathbf{C}) \text{ under } \mathrm{do}(\mathbf{x}) \quad \Longleftarrow \quad (\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C}, \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}},
$$

equivalently $(\mathbf{A} \perp_d \mathbf{B} \mid \mathbf{C})$ in $\mathcal{G} \setminus \mathbf{X}$ (the graph with $\mathbf{X}$ deleted) when $\mathbf{X}$ is disjoint from $\mathbf{A}, \mathbf{B}, \mathbf{C}$. For example, with $A \to X \to D$, the intervention severs A from X, and $\mathbf{X}$ fixed blocks every path through X.

3.1.3.2 Insight 1: adding/removing observations

Graph: $Z \to X \to Y$, $X \leftrightarrow Y$ (via U). Then

$$P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, u_Y, u) P(\mathbf{u}) = P(z) \sum_{u} P(y \mid x, u) P(u).$$

Marginalizing, $P(z \mid \mathrm{do}(x)) = P(z) \sum_u P(u) \sum_y P(y \mid x, u) = P(z)$, so

$$P(y \mid \mathrm{do}(x), z) = \frac{P(y, z \mid \mathrm{do}(x))}{P(z \mid \mathrm{do}(x))} = \sum_{u} P(y \mid x, u) P(u) = \sum_{z} P(y, z \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x)).$$

The observation z can be dropped, and the graphical reason is $(Z \perp_d Y \mid X)_{\mathcal{G}_{\overline{X}}}$. In the original model Z and Y are inseparable: $Z \not\perp\!\!\!\perp Y$ (through $Z \to X \to Y$) and $Z \not\perp\!\!\!\perp Y \mid X$ (X is a collider on $Z \to X \leftrightarrow Y$). In the $\mathrm{do}(X)$-world $\mathcal{M}_x$, however, the edge $Z \to X$ is cut and Y, Z are d-separated; since $P(y \mid \mathrm{do}(x), z) = P_x(y \mid z)$ is an ordinary conditional in the submodel, the observation can be removed.

3.1.3.3 Insight 2: exchanging action and observation

Graph: $Z \to X$, $Z \to Y$, $X \to Y$. After observing Z, Y reacts to X in the same way with and without intervention: given Z, Y is associated with X only through causal paths, so *seeing* $X = x$ is equivalent to *doing* $X = x$. If Z blocks all back-door paths, the remaining association is causation:

$$(Y \perp_d X \mid Z)_{\mathcal{G}_{\underline{X}}} \implies P(y \mid \mathrm{do}(x), z) = P(y \mid x, z).$$

Check: $P(y, z \mid \mathrm{do}(x)) = \sum_{\mathbf{u}} P(z \mid u_Z) P(y \mid x, z, u_Y) P(\mathbf{u}) = P(z) P(y \mid x, z)$ and $P(z \mid \mathrm{do}(x)) = \sum_y P(z) P(y \mid x, z) = P(z)$, so $P(y \mid \mathrm{do}(x), z) = P(z) P(y \mid x, z) / P(z) = P(y \mid x, z)$.

3.1.3.4 Insight 3: adding/removing actions

Graph: $X \to Y$, $Z \leftrightarrow Y$ (latent U_{zy}), $Z \leftrightarrow X$ (latent U_{zx}). If there is no causal path from X to Z, intervening on X has no effect on Z:

$$(Z \perp_d X)_{\mathcal{G}_{\overline{X}}} \implies P(z \mid \mathrm{do}(x)) = P(z).$$

Check: $P(z \mid \mathrm{do}(x)) = \sum_y \sum_{u_{zy}, u_{zx}} P(z \mid u_{zy}, u_{zx}) P(y \mid x, u_{zy}) P(u_{zy}, u_{zx}) = \sum_{u_{zy}, u_{zx}} P(z \mid u_{zy}, u_{zx}) P(u_{zy}, u_{zx}) = P(z)$.

3.1.3.5 The rules

Notation: $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ is $\mathcal{G}$ with edges into $\mathbf{X}$ and out of $\mathbf{Z}$ removed.

Theorem 3.2 (Rules of do-calculus; Pearl 1995)

For any do-distribution induced by a causal model $\mathcal{M}$ with diagram $\mathcal{G}$, and disjoint $\mathbf{X}, \mathbf{Y}, \mathbf{Z}, \mathbf{W}$:

Rule 1 (adding/removing observations) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Rule 2 (action/observation exchange) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$.

Rule 3 (adding/removing actions) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$ if $(\mathbf{Z} \perp_d \mathbf{Y} \mid$

$\mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}}$, where $\mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$.

Proof [Supplement]. Work in the post-intervention model $\mathcal{M}_{\mathbf{x}}$, whose diagram is $\mathcal{G}_{\overline{\mathbf{X}}}$; by the generalized truncated factorization its distribution (with the latent variables written as explicit nodes) is Markov relative to $\mathcal{G}_{\overline{\mathbf{X}}}$, so d-separation in $\mathcal{G}_{\overline{\mathbf{X}}}$ implies conditional independence in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$.

Rule 1. $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$ gives $\mathbf{Y} \perp\!\!\!\perp \mathbf{Z} \mid \mathbf{W}$ in $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ (the variables $\mathbf{X}$ are constants there), which is the claim.

Augmented graph. For Rules 2 and 3 add to $\mathcal{G}_{\overline{\mathbf{X}}}$ a parent F_Z for each $Z \in \mathbf{Z}$ (an "intervention switch"): F_Z = idle leaves the mechanism of Z intact, $F_Z = z'$ replaces it by the constant z'. Then $P(\cdot \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}))$ is the law given $F_{\mathbf{Z}} = \mathbf{z}$, and $P(\cdot \mid \mathrm{do}(\mathbf{x}))$ the law given $F_{\mathbf{Z}} =$ idle. The augmented model is still Markov relative to the augmented graph $\mathcal{G}^F$.

Rule 2. On the event $\mathbf{Z} = \mathbf{z}$, the two regimes $F_{\mathbf{Z}} = \mathbf{z}$ and $F_{\mathbf{Z}} =$ idle differ only in the mechanism of $\mathbf{Z}$, so it suffices that $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z to $\mathbf{Y}$ enters Z through $F_Z \to Z$. If it continues along an edge into Z, then Z is a collider on it with Z in the conditioning set, but the remainder of the path, read from Z, starts with an arrow into Z; if it continues along an edge out of Z, Z is a non-collider in the conditioning set and the path is blocked. Hence the open paths from $F_{\mathbf{Z}}$ are exactly the extensions $F_Z \to Z \leftarrow \cdots$ of paths that leave Z by an incoming edge, and these are open given $(\mathbf{X}, \mathbf{Z}, \mathbf{W})$ iff the corresponding path from Z to $\mathbf{Y}$ is open in $\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}$ (where the outgoing edges of $\mathbf{Z}$ are removed). So $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$ implies $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{Z}, \mathbf{W})_{\mathcal{G}^F}$, which gives Rule 2.

Rule 3. Now $\mathbf{Z}$ is not observed and we need $\mathbf{Y} \perp\!\!\!\perp F_{\mathbf{Z}} \mid \mathbf{X}, \mathbf{W}$ in $\mathcal{G}^F$. A path from F_Z enters Z and continues either out of Z (then Z is an unconditioned non-collider) or into Z (then Z is a collider, open only if Z has a descendant in $\mathbf{W}$, i.e. $Z \in An(\mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}}}$). For $Z \in \mathbf{Z(W)}$ the second kind is blocked, and the first kind is open given $(\mathbf{X}, \mathbf{W})$ exactly when the corresponding path from Z is open in a graph where Z has no incoming edges, i.e. in $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}$. For $Z \in \mathbf{Z} \setminus \mathbf{Z(W)}$, the separation $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z(W)}}}$ blocks the paths through incoming edges as well. Hence the stated condition implies $(F_{\mathbf{Z}} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}^F}$, and the two regimes give the same $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w})$. ■

Unconditional versions ($\mathbf{W} = \emptyset$) read: Rule 1 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}}}$, Rule 2 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{X}}\underline{\mathbf{Z}}}}$, Rule 3 needs $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X})_{\mathcal{G}_{\overline{\mathbf{XZ}}}}$.

Simplified form. Let $\mathcal{H} = \mathcal{G} \setminus \mathbf{X}$ and $Q = P_{\mathbf{x}}$. Then

$$
\begin{aligned}
&\text{R1: } Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}},\\
&\text{R2: } Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{z}, \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\underline{\mathbf{Z}}}},\\
&\text{R3: } Q(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{w}) = Q(\mathbf{y} \mid \mathbf{w}) && \text{if } (\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{W})_{\mathcal{H}_{\overline{\mathbf{Z(W)}}}},\ \mathbf{Z(W)} = \mathbf{Z} \setminus An(\mathbf{W})_{\mathcal{H}}.
\end{aligned}
$$

Every rule is a statement about one fixed submodel $\mathcal{M}_{\mathbf{x}}$; $\mathrm{do}(\mathbf{x})$ is just carried along.

3.1.3.6 Understanding Rule 2

One may add the observation $\mathbf{x}$ under $\mathrm{do}(\mathbf{x})$: $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{x}, \mathbf{z}, \mathbf{w})$. So Rule 2 asks whether the model and the submodel generate the same conditional probability, e.g. (with no outer $\mathrm{do}(\mathbf{x})$) $P(\mathbf{y} \mid \mathrm{do}(\mathbf{z}), \mathbf{z}, \mathbf{w}) = P(\mathbf{y} \mid \mathbf{z}, \mathbf{w})$.

Regime indicator. Augment $\mathcal{G}$ (here $W \to Z$, $W \to Y$, $Z \to Y$) with a parent $F_Z \to Z$ selecting the mechanism:

$$
f'_Z = \begin{cases} f_Z(W, U_Z) & \text{if } F_Z = 0,\\ z & \text{if } F_Z = 1. \end{cases}
$$

Then the Rule 2 equality $P(y \mid \mathrm{do}(z), z, w) = P(y \mid z, w)$ reads $P(y \mid z, w, F_Z = 1) = P(y \mid z, w, F_Z = 0)$, i.e. $(F_Z \perp\!\!\!\perp Y \mid Z, W)$ in the augmented graph $\mathcal{G}'$. Since Z is given, every d-connecting path from F_Z must be $F_Z \to Z \leftarrow \cdots Y$ (the collider Z is opened); so we check paths that leave Z through an arrowhead, i.e. back-door paths, which is the same as checking Z versus Y in $\mathcal{G}_{\underline{Z}}$.

3.1.3.7 Understanding Rule 3

"Changing a function" subsumes "changing it to a constant", so

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathrm{do}(\mathbf{z}), \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), \mathbf{w}) \iff P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), F_Z = 1, \mathbf{w}) = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}), F_Z = 0, \mathbf{w}),$$

which holds if $F_{\mathbf{Z}} \perp\!\!\!\perp \mathbf{Y} \mid \mathbf{W}, \mathbf{X}$ in $\mathcal{G}'$. A path leaving F_Z has the form $F_Z \to Z \to \cdots$ (always feasible) or $F_Z \to Z \leftarrow \cdots$, feasible only if the collider Z is opened, i.e. $Z \in An(\mathbf{W})$ in $\mathcal{G}_{\overline{\mathbf{X}}}$. Hence edges into Z can be cut exactly for $Z \in \mathbf{Z}(\mathbf{W})$, giving the condition $(\mathbf{Z} \perp_d \mathbf{Y} \mid \mathbf{X}, \mathbf{W})_{\mathcal{G}_{\overline{\mathbf{X}}\,\overline{\mathbf{Z}(\mathbf{W})}}}$.

Example 3.1 (Why Z(W) is needed)

Graph: $T \to Z \to Y$, $Z \to W$. Is $P(t \mid \mathrm{do}(z), w) = P(t \mid w)$? In $\mathcal{G}_{\overline{Z}}$, T is isolated, so a naive check $(T \perp_d Z \mid W)_{\mathcal{G}_{\overline{Z}}}$ would say yes. But $Z \in An(W)$, so $\mathbf{Z}(\mathbf{W}) = \emptyset$ and the correct graph is $\mathcal{G}$ itself, where $T \to Z$ remains and the condition fails. Indeed, in almost every compatible model,

$$P(t \mid \mathrm{do}(z), w) = \frac{P(t, w \mid \mathrm{do}(z))}{P(w \mid \mathrm{do}(z))} = \frac{\sum_y P(t)P(w \mid z)P(y \mid z)}{\sum_{y,t} P(t)P(w \mid z)P(y \mid z)} = \frac{P(t)P(w \mid z)}{P(w \mid z)} = P(t) \neq P(t \mid w),$$

since W carries information about T through Z observationally.

3.1.4 Do-calculus and identification

Motivation. State completeness, work the front-door derivation, and give tools for proving non-identifiability.

Theorem 3.3 (Completeness; Huang and Valtorta 2006, Shpitser and Pearl 2006)

$Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $P(\mathbf{v})$ and $\mathcal{G}$ iff there is a finite sequence of applications of the rules of do-calculus and the probability axioms that reduces Q to a do-free expression.

Proof [Supplement]. *If:* each rule of do-calculus and each probability axiom preserves equality for every model compatible with $\mathcal{G}$, so a reduction to a do-free expression writes Q as a functional of $P(\mathbf{v})$, which is identifiability. *Only if:* if Q is identifiable, the ID algorithm (Chapter 3, Section 3.2) returns an expression; each of its steps is an application of the rules of do-calculus and the probability axioms (Huang and Valtorta 2006; Shpitser and Pearl 2006), and when the algorithm fails it exhibits a hedge, from which two models agreeing on $P(\mathbf{v})$ but not on Q are built. The construction of these models is not reproduced here. ■

Syntactic goal: re-express Q without do.

3.1.4.1 Front-door by do-calculus

Smoking $S \to$ Tar $T \to$ Cancer C, with $S \leftrightarrow C$.

$$P(c \mid \mathrm{do}(s)) = \sum_t P(c \mid \mathrm{do}(s), t)P(t \mid \mathrm{do}(s)) \qquad \text{probability axioms}$$

$$
\begin{aligned}
&= \sum_t P(c \mid \mathrm{do}(s), \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\overline{S}\underline{T}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid \mathrm{do}(s)) && \text{Rule 3: } (S \perp_d C \mid T)_{\mathcal{G}_{\overline{T}\,\overline{S}}} \\
&= \sum_t P(c \mid \mathrm{do}(t)) P(t \mid s) && \text{Rule 2: } (T \perp_d S)_{\mathcal{G}_{\underline{S}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', \mathrm{do}(t)) P(s' \mid \mathrm{do}(t)) && \text{probability axioms} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s' \mid \mathrm{do}(t)) && \text{Rule 2: } (T \perp_d C \mid S)_{\mathcal{G}_{\underline{T}}} \\
&= \sum_t P(t \mid s) \sum_{s'} P(c \mid s', t) P(s') && \text{Rule 3: } (T \perp_d S)_{\mathcal{G}_{\overline{T}}}.
\end{aligned}
$$

Justifications: in $\mathcal{G}_{\overline{S}\underline{T}}$ the only edges are $S \leftrightarrow C$ and $S \to T$, and S is given; in $\mathcal{G}_{\overline{T}\,\overline{S}}$, S is disconnected except through $S \leftrightarrow C$, which T does not open; in $\mathcal{G}_{\underline{S}}$, $S \to T$ is removed and $S \leftrightarrow C \leftarrow T$ is blocked at the collider C; in $\mathcal{G}_{\underline{T}}$, $T \leftarrow S \leftrightarrow C$ is blocked by S; in $\mathcal{G}_{\overline{T}}$, T has no edges to S except via the collider C.

3.1.4.2 Proving non-identifiability

Example 3.2

Bow graph $X \to Y$, $X \leftrightarrow Y$, U a fair coin, $\mathcal{F}$: $X \leftarrow U$, $Y \leftarrow f_Y(X, U)$. Define two models that agree with $\mathcal{F}$ whenever $U = X$ (the only case ever observed) but differ off the diagonal:

$$
\mathcal{F}_x^{(1)} : \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 0 & \text{otherwise} \end{cases},
$$

$$
\mathcal{F}_x^{(2)} : \; Y \leftarrow \begin{cases} f_Y(X,U) & U = X \\ 1 & \text{otherwise.} \end{cases}
$$

Under $\mathrm{do}(X = x)$, U and X need not match:

$$
P^{(i)}(Y = 1 \mid \mathrm{do}(x)) = \sum_u P^{(i)}(y \mid x, u) P(u) = P(y \mid x) P(x) + \mathbb{1}_{i=2} \, (1 - P(x)).
$$

Both models induce $\mathcal{G}$ and the same $P(\mathbf{v})$, but the effects differ whenever $P(x) < 1$.

Lemma 3.4 (Graph–subgraph; Tian and Pearl 2002)

If $Q = P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is not identifiable in $\mathcal{G}$, it is not identifiable in any graph obtained by adding a directed or bidirected edge to $\mathcal{G}$. Conversely, if Q is identifiable in $\mathcal{G}$, it remains identifiable after removing edges.

Proof [Supplement]. Let $\mathcal{G}^+$ be $\mathcal{G}$ with an added edge. Every model compatible with $\mathcal{G}$ is compatible with $\mathcal{G}^+$: a structural function may ignore the new parent and an added bidirected edge may correspond to a latent variable with no effect. If Q is not identifiable in $\mathcal{G}$, there are two models compatible with $\mathcal{G}$ with the same $P(\mathbf{v})$ and different Q; they are also compatible with $\mathcal{G}^+$, so Q is not identifiable in $\mathcal{G}^+$. The converse statement is the contrapositive applied to $\mathcal{G}^- \subseteq \mathcal{G}$. ■

(Adding an edge only enlarges the class of compatible models, since a graph permits but does not force dependence.)

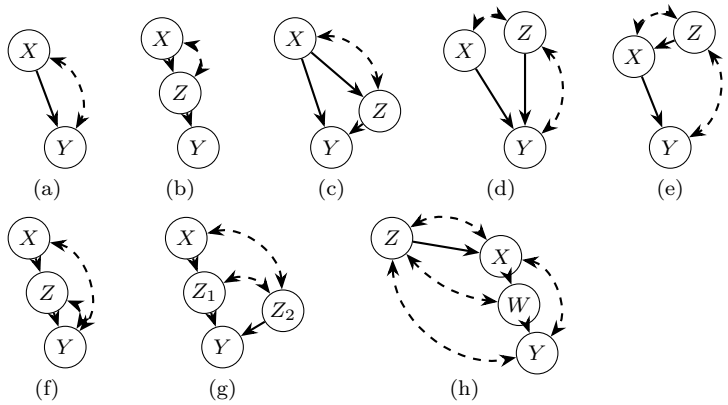
Theorem 3.5 (Non-identifiability criterion, singleton; Tian and Pearl 2002)

If there is a bidirected path connecting X to any of its children in $\mathcal{G}$, then $P(\mathbf{v}' \mid \mathrm{do}(x))$ is not identifiable from $P(\mathbf{v})$ and $\mathcal{G}$.

Proof [Supplement]. *Base case.* If $X \to Y$ and $X \leftrightarrow Y$, take binary U, X, Y. Model 1: $X = U$, $Y = X$. Model 2: $X = U$, $Y = U$. Both give $P(x, y) = P(X = Y = x) = P(U = x)$, but $P(Y = 1 \mid \mathrm{do}(X = 1))$ is 1 in model 1 and $P(U = 1)$ in model 2. *General case.* If a bidirected path $X \leftrightarrow W_1 \leftrightarrow \cdots \leftrightarrow W_k \leftrightarrow Y$ connects X to a child Y, extend the construction: let each latent variable on the path equal a common binary U and let every W_j copy U, while all other variables are independent noise. The two models above, embedded in this structure, still agree on $P(\mathbf{v})$ and disagree on $P(\mathbf{v}' \mid \mathrm{do}(x))$; by the graph–subgraph lemma, adding the remaining edges of $\mathcal{G}$ preserves non-identifiability. (This is the argument of Tian and Pearl 2002.) ■

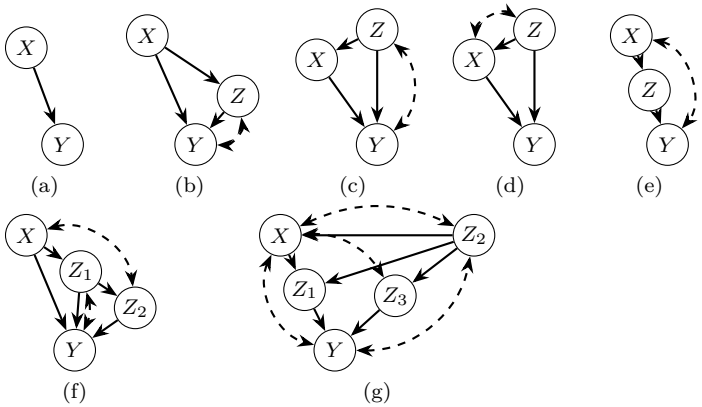
3.1.4.3 Catalogue of graphs

Graphs in which $P(y \mid \mathrm{do}(x))$ is not identifiable (Pearl 2000, Fig. 3.9):



(a) is the bow graph; (b), (c) contain the bow $X \to Z$, $X \leftrightarrow Z$ on the way to Y; in (d)–(h) X is connected to its child by a bidirected path (e.g. in (d) $X \leftrightarrow Z \leftrightarrow Y$, in (g) $X \leftrightarrow Z_2 \leftrightarrow Z_1$ with Z_1 a child of X).

Graphs in which $P(y \mid \mathrm{do}(x))$ is identifiable (Pearl 2000, Fig. 3.8):



For instance in (b) X has no incoming edges, so $P(y \mid \mathrm{do}(x)) = P(y \mid x)$;

(c) and (d) are identified by back-door adjustment for Z (in (d) the confounding between X and Z is harmless), and (e) by the front-door formula.

3.1.5 Back to Simpson's paradox

Motivation. Use the do-operator to dissolve the paradox and explain why statistics alone cannot resolve it.

3.1.5.1 Pearson's shock

Pearson, Lee and Bramley-Moore (1899) observed that mixing heterogeneous groups, each of which shows no "organic" correlation, can produce a correlation, which they called *spurious*; since no community is perfectly homogeneous, correlations are always liable to an error of unknown size, a "shock" to those who equate correlation with causation. Within each subgroup A and B may be uncorrelated, yet in the mixture a correlation appears: an artefact of heterogeneity. The same data can lead to opposite conclusions depending on whether we aggregate or stratify. This is the essence of Simpson's paradox, and it cannot be resolved by statistics alone; it requires causal reasoning.

Examples. (1) UC Berkeley's alleged sex bias in graduate admissions (Bickel et al., *Science* 1975): overall a higher admission rate among male applicants (44% vs. 30%), but within departments a slight bias in favor of women. Women tended to apply to more competitive departments, a societal pattern rather than discrimination by committees. But is conditioning on department the right thing to do? That depends on the causal model, e.g. on latent confounding between department choice and admission.

(2) "Reverse regression" (1970s–80s): in salary-discrimination cases, should one compare salaries of equally qualified men and women, or qualifications of equally paid men and women? (Pearl 2009, §4.5; Bhadane et al. 2025.)

3.1.5.2 Dissolving the paradox with the do-operator

The tables of Chapter 1 give

$$P(Y \mid F, X) < P(Y \mid F, \neg X), \qquad P(Y \mid \neg F, X) < P(Y \mid \neg F, \neg X), \qquad P(Y \mid X) > P(Y \mid \neg X).$$

If instead we have, in do-notation,

$$P(Y \mid F, \mathrm{do}(X)) < P(Y \mid F, \mathrm{do}(\neg X)), \qquad P(Y \mid \neg F, \mathrm{do}(X)) < P(Y \mid \neg F, \mathrm{do}(\neg X)),$$

and the action does not affect F (sex), then necessarily $P(Y \mid \mathrm{do}(X)) < P(Y \mid \mathrm{do}(\neg X))$. Proof:

$$P(y \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f \mid \mathrm{do}(x)) = \sum_f P(y \mid \mathrm{do}(x), f) P(f),$$

by Rule 3 (F is not a descendant of X). Both sides are averages of the stratum-specific effects with the *same* weights $P(f)$, so the inequality is preserved.

3.1.5.3 The paradox stems from causal interpretation

Surprise surfaces only when we speak about efficacy, not about evidence for prediction. When two causal models generate the same data and one recommends the drug while the other does not, the decision must be driven by causal considerations; no statistical criterion warns us against consulting the wrong table.

Theorem 3.6 (Sure-thing principle; Savage 1972, Pearl 2000 Thm. 6.1.1)

An action X that increases the probability of an event Y in each subpopulation must also increase the probability of Y in the population as a whole, provided that the action does not change the distribution of the subpopulations.

Proof [Supplement]. Let F index the subpopulations, with $P(f \mid \mathrm{do}(x)) = P(f \mid \mathrm{do}(\neg x)) = P(f)$ by assumption. Then

$$P(Y \mid \mathrm{do}(x)) = \sum_f P(Y \mid \mathrm{do}(x), f)P(f) > \sum_f P(Y \mid \mathrm{do}(\neg x), f)P(f) = P(Y \mid \mathrm{do}(\neg x)),$$

since the inequality holds term by term and the weights $P(f)$ are the same on both sides. The reversal in Simpson's paradox is impossible here because, unlike conditional proportions, the two sides are averaged with identical weights. ■

Any statistical relationship between two variables can be reversed by including additional factors in the analysis; people think in terms of causes, not proportions. Reversal is possible in the calculus of proportions (prediction), impossible in the calculus of causation. Applications include the adjustment problem, sex and salary discrimination cases, the design of optimal treatments and fairness in machine learning.

3.1.5.4 Appendix: the Berkeley data

Department	Men applied	Men admitted	Women applied	Women admitted
A	825	62%	108	82%
B	560	56%	25	68%
C	325	37%	593	34%
D	417	33%	375	35%
E	191	28%	393	24%
F	373	6%	341	7%
Total	2691	44%	1835	30%

Women applied disproportionately to competitive departments (C–F). Which causal graph is appropriate?

- $\mathcal{G}_1$: Sex $\to$ Dept $\to$ Admit, Sex $\to$ Admit (Dept is a mediator). The total effect $P(\text{admit} \mid \text{do(sex)})$ is the aggregate rate; adjusting for Dept yields only the (controlled) direct effect. The relevant questions become: *direct effect*, is there discrimination within departments? *indirect effect*, does sex influence which department one applies to?
- $\mathcal{G}_2$: $U \to$ Dept, $U \to$ Sex, Dept $\to$ Admit, Sex $\to$ Admit, with U latent (Dept is a confounder via U). Then Dept blocks the back-door path Sex $\leftarrow U \to$ Dept $\to$ Admit, and back-door adjustment gives $P(\text{admit} \mid \text{do(sex)}) = \sum_d P(\text{admit} \mid \text{sex}, d)P(d)$, which reveals no significant discrimination within departments.

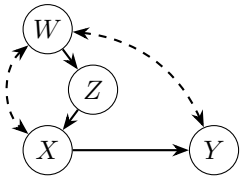
Same data, same adjustment, but different causal interpretations.

3.1.6 Summary and food for thought

- Do-calculus gives a syntactic characterization of policy evaluation for atomic interventions.
- The problem of confounding and identification is essentially solved non-parametrically.
- Simpson's paradox is mathematised and dissolved.

- Applications pervade the social and health sciences, statistics, ML and AI.

Food for thought: the napkin graph. $W \to Z \to X \to Y$, $W \leftrightarrow X$, $W \leftrightarrow Y$. Is $P(y \mid \mathrm{do}(x))$ identifiable?



Yes, although neither back-door nor front-door applies. Intervening on Z makes X an unconfounded treatment in the subpopulation defined by W:

$$P(y \mid \mathrm{do}(x)) = \frac{\sum_w P(x, y \mid z, w)P(w)}{\sum_w P(x \mid z, w)P(w)} \qquad \text{(for any } z\text{)}.$$

(Derivation: $P(y \mid \mathrm{do}(x)) = P(y \mid \mathrm{do}(x), \mathrm{do}(z))$ by Rule 3; $P(x, y \mid \mathrm{do}(z)) = \sum_w P(x, y \mid z, w)P(w)$ by back-door through W; and $P(y \mid \mathrm{do}(z), \mathrm{do}(x)) = P(y \mid \mathrm{do}(z), x)$ by Rule 2, since in $\mathcal{G}_{\overline{Z}\underline{X}}$ the only paths from X to Y go through W, which is cut from X only via Z. Section 3.2 obtains this systematically.)

3.2 Algorithmic identification

Motivation. The central question is how to decide identifiability of $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ mechanically, without searching for a clever sequence of do-calculus steps.

Roadmap. (1) Define a factorization of the distributions generated by an SCM based on its causal diagram (c-components and c-factors).

(2) Establish operations that identify particular factors from a distribution.

(3) Express the target effect in terms of factors and identify each one independently.

3.2.1 Factorizing observational distributions

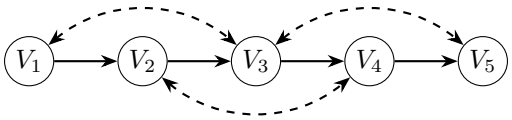
Motivation. Group variables that share latent confounders and show that the joint distribution factorizes over these groups.

3.2.1.1 Confounded components

Definition (C-component)

When V_i and V_j share a common unobservable parent U, their mechanisms are tied together through U, and V_i, V_j are in the same *confounded component* (c-component). The relation is closed under chaining: two variables are in the same c-component iff they are connected by a path of bidirected edges.

Running example: $V_1 \to V_2 \to V_3 \to V_4 \to V_5$ with U_1: $V_1 \leftrightarrow V_3$, U_2: $V_3 \leftrightarrow V_5$, U_3: $V_2 \leftrightarrow V_4$.



V_1 is in the same c-component as V_3, V_3 as V_5, hence V_1 as V_5; V_2 is with V_4. To see this easily, look at the graph induced by the bidirected edges only. The c-component relation is reflexive, symmetric and transitive, so it *partitions* the observed variables: here $\{V_1, V_3, V_5\}$ and $\{V_2, V_4\}$.

3.2.1.2 Markovian case

$$P(\mathbf{v}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid v_1, \ldots, v_{i-1}, \mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i).$$

For $Z \to X$, $Z \to Y$, $X \to Y$:

$$\begin{aligned} P(z, x, y) &= \sum_{u_z} P(z \mid u_z) P(u_z) \sum_{u_x} P(x \mid z, u_x) P(u_x) \sum_{u_y} P(y \mid x, z, u_y) P(u_y) \\ &= P(z) P(x \mid z) P(y \mid x, z). \end{aligned}$$

The factors $P(v_i \mid \mathbf{pa}_i)$ are "canonical", and every one of them is computable from $P(\mathbf{v})$:

$$P(v_i \mid \mathbf{pa}_i) = \frac{\sum_{\mathbf{v} \setminus (\{v_i\} \cup \mathbf{pa}_i)} P(\mathbf{v})}{\sum_{\mathbf{v} \setminus \mathbf{pa}_i} P(\mathbf{v})},$$

$$\text{e.g. } P(z) = \sum_{x,y} P(\mathbf{v}),\ P(x \mid z) = \frac{\sum_y P(\mathbf{v})}{\sum_{x,y} P(\mathbf{v})},\ P(y \mid x, z) = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})}.$$

3.2.1.3 Semi-Markovian case

With only U_1 ($V_1 \leftrightarrow V_3$) and U_2 ($V_3 \leftrightarrow V_5$):

$$P(\mathbf{v}) = P(v_2 \mid v_1) P(v_4 \mid v_3) \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2).$$

Adding U_3 ($V_2 \leftrightarrow V_4$):

$$\begin{aligned} P(\mathbf{v}) = &\Big[\sum_{u_3} P(v_2 \mid v_1, u_3) P(v_4 \mid v_3, u_3) P(u_3) \Big] \\ &\cdot \Big[\sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2) \Big]. \end{aligned}$$

The sums separate along c-components because different c-components share no latent variable.

3.2.1.4 C-factors

These factors are long to write in terms of $P(\mathbf{v}, \mathbf{u})$, but their structure follows from the diagram. Define for $C \subseteq \mathbf{V}$

$$Q[C](\mathbf{c}, \mathbf{pa}_C) = \sum_{\mathbf{u}(C)} P(\mathbf{u}(C)) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i),$$

$$\mathbf{U}(C) = \bigcup_{V_i \in C} \mathbf{U}_i.$$

Then

$$P(\mathbf{v}) = Q[V_2, V_4](v_2, v_4, v_1, v_3) \cdot Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4),$$

abbreviated $P(\mathbf{v}) = Q[V_1, V_3, V_5]\, Q[V_2, V_4]$, without naming $\mathbf{U}$ at all. Note $Q[\mathbf{V}] = P(\mathbf{v})$, and by convention $Q[\emptyset] = 1$.

3.2.1.5 C-factors are causal effects

For $C \subseteq \mathbf{V}$, consider the effect of all other variables on C. By the truncated factorization,

$$P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) = \sum_{\mathbf{u}(C)} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u}(C)) = Q[C],$$

since latent variables that are not parents of C sum out. *C-factors are interventional distributions*; this is the key link between factorization and identification. For example $Q[V_1, V_3, V_5](v_1, v_3, v_5, v_2, v_4) = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) P(v_3 \mid v_2, u_1, u_2) P(v_5 \mid v_4, u_2)$ is the distribution of (V_1, V_3, V_5) when $V_2 = v_2$ and $V_4 = v_4$ are fixed by intervention.

3.2.2 C-factor algebra

Motivation. Develop the two operations, marginalizing to ancestral sets and factorizing into c-components, that drive identification.

3.2.2.1 Marginalizing variables in c-factors

C-factors behave partly like probabilities. In $Q[V_1, V_3, V_5]$ each of V_1, V_3, V_5 appears in a single term (none is a parent of another member), so summing over v_3 gives

$$\sum_{v_3} Q[V_1, V_3, V_5] = \sum_{u_1, u_2} P(u_1, u_2) P(v_1 \mid u_1) \Big(\sum_{v_3} P(v_3 \mid v_2, u_1, u_2) \Big) P(v_5 \mid v_4, u_2) = Q[V_1, V_5].$$

In $Q[V_1, V_2, V_3]$, by contrast, V_1 is a parent of V_2. Summing over v_1:

$$\sum_{v_1} Q[V_1, V_2, V_3] = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) \sum_{v_1} P(v_1 \mid u_1) P(v_2 \mid v_1, u_3),$$

whereas

$$Q[V_2, V_3](v_2, v_3, v_1) = \sum_{\mathbf{u}} P(\mathbf{u}) P(v_3 \mid v_2, u_1, u_2) P(v_2 \mid v_1, u_3)$$

keeps v_1 as a fixed argument. The coupling of v_1 with u_1 (shared with V_3) inside the sum prevents the reduction, so $\sum_{v_1} Q[V_1, V_2, V_3] \neq Q[V_2, V_3]$ in general: one can only sum out variables that do not feed into the remaining ones.

Lemma 3.7 (Ancestral reduction)

Let $W \subseteq C \subseteq \mathbf{V}$. If W is ancestral in $\mathcal{G}[C]$ (it contains all its ancestors within the subgraph induced by C), then

$$Q[W] = \sum_{\mathbf{c} \setminus \mathbf{w}} Q[C].$$

Proof [Supplement]. Order C topologically and sum out the variables of $C \setminus W$ one at a time from the last. When $V \in C \setminus W$ is summed, it has no child left among the remaining variables: a child in W would make V an ancestor of W in $\mathcal{G}[C]$, hence $V \in W$ since W is ancestral; children in $C \setminus W$ come later in the order and were already summed out. So v appears in $Q[C] = \sum_{\mathbf{u}} \prod_{V_i \in C} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$ only in its own factor, which sums to one; the latent variables attached only to summed variables are then integrated out as well. What remains is $Q[W]$. ■

Example: For $C = \{V_1, V_2, V_3\}$ in $\mathcal{G}[C]$ ($V_1 \to V_2 \to V_3$): $\{V_1, V_2\}$ and $\{V_1\}$ are ancestral, $\{V_2, V_3\}$ is not.

3.2.2.2 C-component factorization

Lemma 3.8 (Tian 2002)

For any $H \subseteq \mathbf{V}$, let $H_1, \dots, H_k$ be the c-components of $\mathcal{G}[H]$. Then

$$Q[H] = \prod_{j=1}^{k} Q[H_j].$$

Moreover, let $V^{(1)} < \cdots < V^{(|H|)}$ be a topological order of $\mathcal{G}[H]$ and $H^{\le i}$ the variables of H up to and including $V^{(i)}$. Then each factor is computable from $Q[H]$:

$$Q[H_j] = \prod_{V^{(i)} \in H_j} \frac{Q[H^{\le i}]}{Q[H^{\le i-1}]}, \qquad Q[H^{\le i}] = \sum_{\mathbf{h} \setminus \mathbf{h}^{\le i}} Q[H].$$

Proof [Supplement]. The latent variables split into groups, one for each c-component, since a latent variable with children in two c-components would join them by a bidirected path. In $Q[H] = \sum_{\mathbf{u}} \prod_{V_i \in H} P(v_i \mid \mathbf{pa}_i, \mathbf{u}_i) P(\mathbf{u})$, group the factors and the latent sum by c-component: the latent variables of H_j appear only in the factors of H_j, and different groups are independent under $P(\mathbf{u})$, so the sum factorizes into $\prod_j Q[H_j]$. For the second part, $H^{\le i}$ is ancestral in $\mathcal{G}[H]$ (all predecessors in a topological order), so $Q[H^{\le i}] = \sum_{h \setminus h^{\le i}} Q[H]$ by ancestral reduction; the ratio $Q[H^{\le i}]/Q[H^{\le i-1}]$ is therefore a function of $Q[H]$, and multiplying these ratios over the variables of H_j reproduces $Q[H_j]$ (each latent group of H_j is integrated exactly once). ■

($H^{\le i}$ is ancestral in $\mathcal{G}[H]$, so $Q[H^{\le i}]$ follows by ancestral reduction.)

Example 3.3 (Factorizing $P(\mathbf{v})$ by c-components)

Graph: $Z_2 \to X$, $Z_2 \to Z_1$, $Z_2 \to Z_3$, $X \to Z_1$, $Z_1 \to Y$, $Z_3 \to Y$, $Z_2 \leftrightarrow X$, $X \leftrightarrow Z_3$, $X \leftrightarrow Y$, $Z_2 \leftrightarrow Y$. The c-components are $\{Z_2, Z_3, X, Y\}$ and $\{Z_1\}$, so

$$Q[\mathbf{V}] = Q[Z_2, Z_3, X, Y]\, Q[Z_1],$$
$$P(\mathbf{v}) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, z_3, x, y)) = P(z_2, z_3, x, y \mid \mathrm{do}(z_1))\, P(z_1 \mid \mathrm{do}(z_2, x)),$$

the last step because only the parents Z_2, X of Z_1 matter (Rule 3).

Example 3.4 (Running example)

With $H = \mathbf{V}$ and the order $V_1 < \cdots < V_5$, $Q[\mathbf{V}] = Q[V_1, V_3, V_5] Q[V_2, V_4]$ and

$$\begin{aligned} Q[V_1, V_3, V_5] &= \frac{Q[V_1]}{Q[\emptyset]} \cdot \frac{Q[V_1, V_2, V_3]}{Q[V_1, V_2]} \cdot \frac{Q[V_1, \dots, V_5]}{Q[V_1, \dots, V_4]} \\ &= \frac{\sum_{v_2, \dots, v_5} P(\mathbf{v})}{\sum_{\mathbf{v}} P(\mathbf{v})} \cdot \frac{\sum_{v_4, v_5} P(\mathbf{v})}{\sum_{v_3, v_4, v_5} P(\mathbf{v})} \cdot \frac{P(\mathbf{v})}{\sum_{v_5} P(\mathbf{v})} \\ &= \frac{P(v_1)}{1} \cdot \frac{P(v_1, v_2, v_3)}{P(v_1, v_2)} \cdot \frac{P(\mathbf{v})}{P(v_1, \dots, v_4)} = P(v_1) P(v_3 \mid v_1, v_2) P(v_5 \mid v_1, \dots, v_4), \\ Q[V_2, V_4] &= \frac{Q[V_1, V_2]}{Q[V_1]} \cdot \frac{Q[V_1, \dots, V_4]}{Q[V_1, V_2, V_3]} = P(v_2 \mid v_1) P(v_4 \mid v_1, v_2, v_3). \end{aligned}$$

In $\mathcal{G}[\{V_2, V_4\}]$ there are no directed edges ($V_2 \leftrightarrow V_4$ only), so both $\{V_2\}$ and $\{V_4\}$ are ancestral there:

$$Q[V_2] = \sum_{v_4} Q[V_2, V_4] = P(v_2 \mid v_1) \sum_{v_4} P(v_4 \mid v_1, v_2, v_3) = P(v_2 \mid v_1),$$

$$Q[V_4] = \sum_{v_2} P(v_2 \mid v_1)P(v_4 \mid v_1, v_2, v_3).$$

3.2.2.3 Visualization

Starting from $Q[\mathbf{V}]$ one can move down a lattice of c-factors: summing out a variable that is "last" (ancestral reduction, e.g. $Q[\mathbf{V}] \to Q[V_{1234}] \to Q[V_{123}] \to Q[V_{12}] \to Q[V_1]$) or splitting along c-components (e.g. $Q[\mathbf{V}] \to Q[V_{135}] \cdot Q[V_{24}]$, $Q[V_{1234}] \to Q[V_{13}] \cdot Q[V_{24}]$, $Q[V_{123}] \to Q[V_{13}] \cdot Q[V_2]$). From $Q[V_{135}]$ every subset is reachable by summation since $\mathcal{G}[\{V_1, V_3, V_5\}]$ has no directed edges, and $Q[V_{15}] = Q[V_1]Q[V_5]$ factorizes because V_1, V_5 are not bidirected-connected inside $\{V_1, V_5\}$. Once a factor is fully factorized it cannot be factorized further; summation is commutative.

Summary of the algebra.

1. *Reduce to an ancestral set:* $Q[W] = \sum_{\mathbf{c}\setminus\mathbf{w}} Q[C]$ if W is ancestral in $\mathcal{G}[C]$.
2. *Factorize into c-components:* $Q[H] = \prod_j Q[H_j]$, with $H_1, \ldots, H_k$ the c-components of $\mathcal{G}[H]$, each computable from $Q[H]$.

These two operations yield an if-and-only-if condition for identifiability.

3.2.3 Causal queries in terms of c-factors: examples

Motivation. Rederive the back-door, front-door and napkin formulas with c-factor algebra.

3.2.3.1 Back-door

$Z \to X$, $Z \to Y$, $X \to Y$ (Markovian): $P(\mathbf{v}) = Q[X]Q[Y]Q[Z]$. In $\mathcal{G}_{\overline{X}}$, $P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$ with

$$Q[Y] = \frac{Q[X,Y,Z]}{Q[X,Z]} = \frac{P(\mathbf{v})}{\sum_y P(\mathbf{v})} = P(y \mid x, z), \qquad Q[Z] = \sum_{x,y} Q[X,Y,Z] = P(z),$$

so $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid z, x)P(z)$.

3.2.3.2 Front-door

$X \to Z \to Y$, $X \leftrightarrow Y$: $P(\mathbf{v}) = Q[X,Y] \cdot Q[Z]$ with (order $X < Z < Y$)

$$Q[X,Y] = \frac{Q[X]}{Q[\emptyset]} \cdot \frac{Q[X,Z,Y]}{Q[X,Z]} = P(x)P(y \mid x, z), \qquad Q[Z] = \frac{Q[X,Z]}{Q[X]} = P(z \mid x).$$

Then

$$P(y \mid \mathrm{do}(x)) = \sum_z P(y, z \mid \mathrm{do}(x)) = \sum_z Q[Y,Z] = \sum_z Q[Y]Q[Z],$$

and since $\{Y\}$ is ancestral in $\mathcal{G}[\{X,Y\}]$ (no directed edge between them), $Q[Y] = \sum_{x'} Q[X,Y] = \sum_{x'} P(x')P(y \mid x', z)$. Hence $P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(x')P(y \mid x', z)$.

3.2.3.3 Napkin

$W_1 \to W_2 \to X \to Y$, $W_1 \leftrightarrow X$, $W_1 \leftrightarrow Y$. Then $P(\mathbf{v}) = Q[W_1, X, Y]\, Q[W_2]$ and

$$P(y \mid \mathrm{do}(x)) = \sum_{w_1,w_2} Q[W_1, W_2, Y] = Q[Y],$$

because $\{Y\}$ is ancestral in $\mathcal{G}[\{W_1, W_2, Y\}]$ (in that subgraph Y has no parents). Can we get $Q[Y]$ from $Q[W_1, X, Y]$, which is computable from $Q[\mathbf{V}]$? With the order $W_1 < W_2 < X < Y$,

$$Q[W_1, X, Y] = \frac{Q[W_1]}{Q[\emptyset]} \cdot \frac{Q[W_1, W_2, X]}{Q[W_1, W_2]} \cdot \frac{Q[W_1, W_2, X, Y]}{Q[W_1, W_2, X]} = P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x).$$

$\{X, Y\}$ is ancestral in $\mathcal{G}[\{W_1, X, Y\}]$, so $Q[X, Y] = \sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)$. In $\mathcal{G}[\{X, Y\}]$, X and Y are no longer bidirected-connected (W_1 is gone), so $Q[X, Y] = Q[X]Q[Y]$, giving

$$P(y \mid \mathrm{do}(x)) = Q[Y] = \frac{Q[X, Y]}{Q[X]} = \frac{Q[X, Y]}{\sum_y Q[X, Y]} = \frac{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)P(y \mid w_1, w_2, x)}{\sum_{w_1} P(w_1)P(x \mid w_1, w_2)}.$$

The same result by do-calculus.

$$\begin{aligned}
P(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x, w_1, w_2)) && \text{Rule 3} \\
&= P(y \mid x, \mathrm{do}(w_1, w_2)) && \text{Rule 2} \\
&= \frac{P(y, x \mid \mathrm{do}(w_1, w_2))}{P(x \mid \mathrm{do}(w_1, w_2))} = \frac{P(y, x \mid \mathrm{do}(w_2))}{P(x \mid \mathrm{do}(w_2))} && \text{Rule 3} \\
&= \frac{\sum_{w_1} P(y, x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))}{\sum_{w_1} P(x \mid \mathrm{do}(w_2), w_1)P(w_1 \mid \mathrm{do}(w_2))} && \\
&= \frac{\sum_{w_1} P(y, x \mid w_2, w_1)P(w_1)}{\sum_{w_1} P(x \mid w_2, w_1)P(w_1)} && \text{Rules 2 and 3.}
\end{aligned}$$

3.2.3.4 Overview of the three derivations

Back-door	Front-door	Napkin
$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[X, Y]\,Q[Z]$	$Q[\mathbf{V}] \xrightarrow{\text{fact.}} Q[W_1, X, Y]\,Q[W_2]$
	$Q[X, Y] \xrightarrow{\text{sum}} Q[Y]$	$Q[W_1, X, Y] \xrightarrow{\text{sum}} Q[X, Y] \xrightarrow{\text{fact.}} Q[X]\,Q[Y]$
$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = \sum_z Q[Y]Q[Z]$	$P(y \mid \mathrm{do}(x)) = Q[Y]$

3.2.4 A general identification algorithm

Motivation. Combine the two operations into a sound and complete procedure.

3.2.4.1 Reducing the query to c-factors

Given $\mathcal{G}$ and $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$, first use Rule 3 to enlarge the intervention: every variable that is not an ancestor of $\mathbf{Y}$ in $\mathcal{G}_{\overline{\mathbf{X}}}$ (non-ancestors of $\mathbf{Y}$, and variables that affect $\mathbf{Y}$ only through $\mathbf{X}$) can be intervened on without changing the query. Let $C = An(\mathbf{Y})_{\mathcal{G}_{\overline{\mathbf{X}}}}$. Then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c} \setminus \mathbf{y}} P(\mathbf{c} \mid \mathrm{do}(\mathbf{v} \setminus \mathbf{c})) = \sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} Q[\mathbf{V} \setminus \mathbf{X}] = \sum_{\mathbf{c} \setminus \mathbf{y}} Q[C].$$

If $\mathcal{G}[C]$ has c-components $C_1, \ldots, C_k$, then

$$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x})) = \sum_{\mathbf{c} \setminus \mathbf{y}} \prod_{i=1}^{k} Q[C_i],$$

while the data give $P(\mathbf{v}) = \prod_j Q[T_j]$ with $T_1, \ldots, T_\ell$ the c-components of $\mathcal{G}$, each $Q[T_j]$ computable from $P(\mathbf{v})$. Each C_i lies inside a single T_j. It remains to decide whether $Q[C_i]$ is computable from $Q[T_j]$.

3.2.4.2 The Identify procedure

Algorithm 1 IDENTIFY$(C, T, Q[T], \mathcal{G})$ (decides whether $Q[C]$ is computable from $Q[T]$, $C \subseteq T$)

$A \leftarrow An(C)_{\mathcal{G}[T]}$
if $A = C$ **then** ▷ C is ancestral in T
  return $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$ ▷ marginalize
else if $A = T$ **then** ▷ nothing to marginalize, unable to factorize
  return FAIL
else
  $A_i \leftarrow$ the c-component of $\mathcal{G}[A]$ containing C
  obtain $Q[A] = \sum_{\mathbf{t}\setminus\mathbf{a}} Q[T]$, then $Q[A_i]$ by c-component factorization ▷ marginalize, then factorize
  return IDENTIFY$(C, A_i, Q[A_i], \mathcal{G})$

The three cases. (1) $A = C$: marginalize, $Q[C] = \sum_{\mathbf{t}\setminus\mathbf{c}} Q[T]$.

(2) $A = T$: C is strictly inside T, but T is its own ancestral closure and a single c-component, so T cannot be shrunk: fail.

(3) $C \subsetneq A \subsetneq T$: marginalize to A, factorize A into c-components $A_i, A_j, \ldots$, keep the one containing C (it must be contained in a single component because C is a c-component of $\mathcal{G}[C]$ and $C \subseteq A_i$), and recurse. Each step strictly shrinks T (recursive shrinking: $T \supsetneq A_i \supsetneq \cdots$), so the procedure terminates in at most $|T|$ steps.

Theorem 3.9 (Completeness; Tian 2002)

$P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$ is identifiable from $\mathcal{G}$ and $P(\mathbf{v})$ if and only if each c-factor $Q[C_i]$ is identified by IDENTIFY$(C_i, T_i, Q[T_i], \mathcal{G})$, where T_i is the c-component of $\mathcal{G}$ containing C_i.

Proof [Supplement]. *If:* each call to IDENTIFY returns an expression in $Q[T_i]$, which is identified from $P(\mathbf{v})$ by the c-component factorization (the lemma above with $H = \mathbf{V}$); combining the factors by the generalized truncated factorization gives $P(\mathbf{y} \mid \mathrm{do}(\mathbf{x}))$. *Only if:* when IDENTIFY fails for some C_i, the failing configuration $A = T$ with $C_i \subsetneq T$ contains a hedge, and Shpitser and Pearl (2006) construct from it two models that agree on $P(\mathbf{v})$ but not on the query; this construction is not reproduced here. ■

See also Huang and Valtorta (2006) and Shpitser and Pearl (2006), who proved completeness of the equivalent ID algorithm and, through it, of do-calculus. The failure case $A = T$ corresponds to a graphical structure (a "hedge") from which two models agreeing on $P(\mathbf{v})$ but not on the query can be built.

Example 3.5 (Bow graph fails)

$X \to Y$, $X \leftrightarrow Y$, query $P(y \mid \mathrm{do}(x))$: $C = An(Y)_{\mathcal{G}_{\overline{X}}} = \{Y\}$, $T = \{X, Y\}$. IDENTIFY$(\{Y\}, \{X, Y\}, \ldots)$: $A = An(Y)_{\mathcal{G}[T]} = \{X, Y\} = T$, so it fails, matching Chapter 2.

Example 3.6 (Front-door succeeds)

$X \to Z \to Y$, $X \leftrightarrow Y$: $C = \{Z, Y\}$ with c-components $\{Z\}$, $\{Y\}$ in $\mathcal{G}[C]$. $Q[Z]$ is itself a c-component of $\mathcal{G}$. For $Q[Y]$ with $T = \{X, Y\}$: $A = An(Y)_{\mathcal{G}[T]} = \{Y\} = C$ (the edge $X \to Z \to Y$ is not inside T), so $Q[Y] = \sum_x Q[X, Y]$, as found above.